PREFACE

Maintaining the reliability, availability, and integrity of online systems is becoming increasingly difficult due to the increasing frequency of Distributed Denial of Service (DDoS) assaults. Traditional techniques of detection and defence are sometimes inadequate to keep up with the increasing sophistication of these attacks. Using a strong and creative machine learning-based approach, this study, "Enhancing DDoS Attack Detection Using Deep Learning and Synthetic Data Augmentation," primarily makes use of deep learning and Generative Adversarial Networks (GANs).

This study's main goals are to: (1) create a robust deep learning framework that can precisely identify DDoS attacks and use cutting-edge techniques to neutralize them; and (2) incorporate GANs to improve the quality of the input data and fortify the models against emerging cyberthreats.

The study's research topic aim to assess the effectiveness of different deep learning techniques within the suggested framework and comprehend how GAN-generated synthetic data affects the efficiency and precision of DDoS attack detection models. The study specifically attempts to determine how much artificial data augmentation-based models beat models trained only on real data, as well as how well these algorithms detect and thwart DDoS attacks.

In order to improve the quality of the input data, this study systematically evaluates three different deep learning algorithms: Deep Belief Network (DBN), Long Short-Term Memory (LSTM), and Convolutional Neural Network (CNN). This assessment's use of sophisticated feature selection methods is one of its main goals. The results of this study should greatly advance the existing state of this field's research and provide network security professionals with fresh knowledge and resources. This study also emphasizes the significance of preserving adaptability in the face of changing cyberthreats, promoting ongoing improvements to deep learning models, investigating cutting-edge feature selection techniques, and consistently raising the caliber of datasets.